

Julia Sets

Recall: • Filled-in Julia set, $K(f) = \{z \in \mathbb{C} : |f^k(z)| \nrightarrow \infty\}$,
 where f is a polynomial of degree $n \geq 2$.
 • Julia set, $J(f) = \partial K(f)$.

• Lemma: Let $f(z)$ be a polynomial of degree ≥ 2 .
 Then $\exists \tau > 0$ s.t. $|z| \geq \tau \Rightarrow |f(z)| \geq 2|z|$.
 In particular, if $|f^m(z)| \geq \tau$ for some $m \geq 0$,
 then $|f^k(z)| \rightarrow \infty$ as $k \rightarrow \infty$.
 Hence, either $|f^k(z)| \rightarrow \infty$ or $\{f^k(z) : k=0,1,2,\dots\}$ is bounded.

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Theorem: Let $f(z)$ be a polynomial of degree ≥ 2 .
 Then, $K(f)$ and $J(f)$ are nonempty compact
 subsets of $\mathbb{C}$ with $J(f) \subseteq K(f)$.
 Also, $J(f)$ has empty interior.

Proof: Let $\tau > 0$ be as in the previous lemma.
 Then, $z \in K(f) \Rightarrow |f^k(z)| \nrightarrow \infty$ as $k \rightarrow \infty$
 $\Rightarrow |z| < \tau$.

$$\therefore K(f) \subseteq B(0, \tau)$$

$\Rightarrow K(f)$ is bounded.

Hence, $J(f) = \partial K(f)$ is also bounded.

To show: $K(f)$ is closed.

Let $z \in \mathbb{C} \setminus K(f)$. Then $|f^k(z)| \rightarrow \infty$ as $k \rightarrow \infty$.
 $\Rightarrow |f^m(z)| > \tau$ for some $m \in \mathbb{N}$.

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By the continuity of f^m , $|f^m(w)| > \delta$
 for all w in an open neighborhood of z .
 $\Rightarrow |f^k(w)| \rightarrow \infty$ as $k \rightarrow \infty$ (by the prev. lemma)
 $\Rightarrow w \in \mathbb{C} \setminus K(f)$.
 $\therefore \mathbb{C} \setminus K(f)$ is open $\Rightarrow K(f)$ is closed.
 Hence, $J(f) \subseteq K(f)$ and $J(f)$ is closed.
 $\therefore$ Both $J(f)$ and $K(f)$ are compact.
 Claim: $K(f) \neq \emptyset$.
 Let $z_0 \in \mathbb{C}$ s.t. $f(z_0) = z_0$
 (such a z_0 exists by the fundamental theorem of algebra).
 Then $f^k(z_0) = z_0 \quad \forall k \Rightarrow z_0 \in K(f)$.

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Now, let $z_1 \in \mathbb{C} \setminus K(f)$.
 Then consider the line segment $(1-\lambda)z_0 + \lambda z_1$,
 $\lambda \in [0, 1]$, joining z_0 and z_1 .
 Let $\lambda_0 = \sup \{ \lambda \in [0, 1] : (1-\lambda)z_0 + \lambda z_1 \in K(f) \}$
 Then $(1-\lambda_0)z_0 + \lambda_0 z_1 \in \partial K(f) = J(f)$ (why?)
 $\therefore J(f) \neq \emptyset$.

Finally, if $J(f)$ has nonempty interior,
 there is a nonempty open set $U \subseteq J(f) \subseteq K(f)$
 $\Rightarrow U$ is in the interior of $K(f)$
 $\Rightarrow U \cap \partial K(f) = \emptyset \Rightarrow U \cap J(f) = \emptyset$
 which is a contradiction since $U \subseteq J(f)$.

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Proposition: The Julia set $J = J(f)$ of f is forward and backward invariant under f , ie. $J = f(J) = \bar{f}^{-1}(J)$.

Proof: Let $z \in J$. Then $|f^k(z)| \rightarrow \infty$ as $k \rightarrow \infty$.
Also, we can find $w_n \in \mathbb{C} \setminus K(f)$ st.

$w_n \rightarrow z$.
Then $|f^k(w_n)| \rightarrow \infty$ as $k \rightarrow \infty$ for each n .
Since $|f^k(z)| \rightarrow \infty$, $|f^k(f(z))| \rightarrow \infty$ as $k \rightarrow \infty$
 $\Rightarrow f(z) \in K(f)$

But $|f^k(f(w_n))| \rightarrow \infty$ as $k \rightarrow \infty \Rightarrow f(w_n) \in \mathbb{C} \setminus K(f)$
and $f(w_n) \rightarrow f(z)$ as $n \rightarrow \infty$.
 $\Rightarrow f(z) \in \mathbb{C} \setminus K(f) \Rightarrow f(z) \in \partial K(f) = J$.

Hence, $f(J) \subseteq J$

$\Rightarrow J \subseteq \bar{f}^{-1}(f(J)) \subseteq \bar{f}^{-1}(J)$

$\therefore f(J) \subseteq J \subseteq \bar{f}^{-1}(J)$

We need to show that $\bar{f}^{-1}(J) \subseteq J$.

Let $z_0 \in \bar{f}^{-1}(J)$. Then $f(z_0) = z \in J$.

Let $\{w_n\}$ be a seq. in $\mathbb{C} \setminus K(f)$

st. $w_n \rightarrow z$.
We choose $\{v_n\}$ st. $f(v_n) = w_n$ and $v_n \rightarrow z_0$. (why?)

Then $|f^k(v_n)| = |f^{k-1}(w_n)| \rightarrow \infty$ as $k \rightarrow \infty$
 $\wedge v_n \in \mathbb{C} \setminus K(f)$

Also, $|f^k(z_0)| = |f^{k-1}(z)| \rightarrow \infty \Rightarrow z_0 \in J$

Proposition : $J(f^p) = J(f)$ for any $p \in \mathbb{N}$,
 where $f^p = \underbrace{f \circ \dots \circ f}_{p \text{ times}}$

Proof : It is enough to show that

$$\mathbb{C} \setminus K(f^p) = \mathbb{C} \setminus K(f) .$$

$$z \in \mathbb{C} \setminus K(f) \Leftrightarrow |f^k(z)| \rightarrow \infty \text{ as } k \rightarrow \infty$$

$$(\text{why?}) \rightarrow \Leftrightarrow |(f^p)^k(z)| = |f^{kp}(z)| \rightarrow \infty \text{ as } k \rightarrow \infty .$$

$$\Leftrightarrow z \in \mathbb{C} \setminus K(f^p) .$$

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